THOMAS YOUSEF

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EDUCATION

Boston University, Faculty of Computing & Data Science — Boston, MA

B.S. in Data Science | Minor in Business Administration — Expected May 2026

GPA: 3.67/4.0

Coursework: Analysis of Variance, Data mechanics, Applied Statistics, Principles of Machine Learning, Cybersecurity

Activities: BU Finance and Investments Club, Kappa Theta Pi Technology Fraternity, Reformed University Fellowship

EXPERIENCE

JP Morgan Chase — Wilmington, DE

Data Science Summer Analyst — June 2025 – Aug 2025

- Built and tuned an **XGBoost** regression model to forecast Chase's total daily customer spend across all debit and credit card transactions nationwide, applying feature engineering and time series techniques
- Achieved **1.5%** and **1.9%** error across two testing rounds, enabling better strategic planning, risk management, and anomaly detection for the firm

Fresenius Medical Care — Lexington, MA

Global IT Enterprise Solutions Intern — June 2024 – Aug 2024

- Developed scalable web components using **React.js** and **Node.js**, and designed backend architecture with **AWS SAM** integrated with **DynamoDB** for efficient data storage
- Executed end-to-end testing to validate functionality, data accuracy, and alignment with business requirements

BU Finance and Investments Club — Boston, MA

Assistant VP of Quantitative Research — Sep 2023 – Present

- Organized lectures for junior developers on complex finance topics including Binomial Options Pricing Models, Modern Portfolio Theory, Monte Carlo Methods, and more

PROJECTS

Boston KTP App | GitHub Link – App Store — May 2024 - Present

- Developed a full stack mobile app for Kappa Theta Pi using **React Native, Node.js**, **Firebase,** and **MongoDB**, serving more than 300+ active users with 1,000+ database invocations daily (REST).
- Integrated **Google OAuth** authentication with **Firebase** and implemented persistent login sessions, eliminating repeated logins while maintaining secure access

Stock Clustering Model | GitHub Link — Sep 2024 – Dec 2024

- Developed a website with **React**, and **Flask** that analyzes users' stock holdings using volatility scoring and historical data to recommend lower-risk, same-sector substitutes and predict future portfolio performance.
- Implemented **K-means clustering** on S&P 500 to suggest similar companies based on financial metrics such as dividend yield, volume, and profit margin, improving investor decision-making and portfolio diversification.

Sports Betting Arbitrage | GitHub Link — Jan 2024 – May 2024

- Developed a **Python** based arbitrage detection system that leveraged the **Odds-API** to aggregate and analyze live betting odds across 15+ sportsbooks and 70 sports, identifying arbitrage opportunities.
- Implemented real-time in-game odds tracking, collecting and comparing sportsbook data every 5 minutes to uncover timing-based arbitrage opportunities and validate discrepancies in live odds updates between platforms.

COVID-19 Policy Efficiency | GitHub Link — Oct 2023 - Dec 2023

- Led a team to create a modern data pipeline solution in **Azure** that extracted, cleaned, loaded and analyzed COVID-19 policy data to discover which policies were most effective in reducing COVID-19 cases.
- Created a **galaxy schema** for data in an **Azure ODS**, which was loaded in an **Azure synapse warehouse** and connected to **PowerBI** for statistical analysis and evaluation.

Sentiment Driven Stock Selector | GitHub Link — May 2023 - Jul 2023

- Engineered a sentiment analysis pipeline with **Python** and **NLTK** to scrape and analyze 100 daily news headlines for each S&P 500 company, identifying correlations between market sentiment and short-term stock movements.
- Discovered **strong positive sentiment–return** correlations in 60+ companies, demonstrating the predictive potential of news sentiment in equity performance.

SKILLS

Technical Skills: Python - Java - HTML/CSS - JavaScript - MATLAB - SQL - R

Software: React/React Native - Node.js - Docker - OpenShift- Azure - AWS - PowerBI

Languages: English – Arabic (Native)